Anthony James Padavano

STAFF SOFTWARE ENGINEER & DURABLE AI SYSTEMS

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LINKS

Portfolio · Omni-Dromenon Engine · ORGANVM System · GitHub

PROFILE

Systems architect building durable agentic loops and self-referential state machines for production-grade AI reliability. Sole architect of a 113-repository system across 8 GitHub organizations — 82K files, 23,470 tests, 104 CI/CD pipelines — governing non-deterministic AI outputs through formal state machines and self-healing orchestration. Practitioner of an AI-conductor methodology: human intelligence directing AI-generated volume through structured review gates. Full-Stack Developer (Meta). 18 years professional experience. MFA Creative Writing. LGBTQ+.

TECHNICAL SKILLS

Python · TypeScript · State Machines · Durable Execution · Claude Agent SDK · Multi-Agent Orchestration · Node.js · YAML/JSON Schema · Docker · Redis · GitHub Actions · CI/CD · pytest · Vitest · Zod · Express · WebSocket · GCP / Terraform · Git · pnpm Monorepos · Nginx · React 18

SELECTED PROJECTS

ORGANVM Eight-Organ System — Durable State Machine Infrastructure
113-repository system across 8 GitHub organizations with formal promotion state machine (LOCAL → GRADUATED). Forward-only progression, validated transitions, 50 dependency edges. 82K files, 23,470 tests, 104 CI/CD pipelines.

agentic-titan — Multi-Agent Orchestration Framework
Stateful agent framework with durable retry logic, checkpointing, memory management, and self-correction hooks. Handles non-deterministic failures: hallucination, context overflow, tool-call loops, partial completions. 1,095+ tests.

Application Pipeline — Durable Workflow CLI Suite
Forward-only state machine (research → staged → outcome) with 3,266 tests. YAML schema validation, multi-model evaluation, Greenhouse API integration. 127 commands across 60+ modules. Python.

Omni-Dromenon Engine — Real-Time Distributed Consensus System
TypeScript pnpm monorepo: Express/Socket.io with weighted consensus, React 18 interfaces, OSC bridge to SuperCollider/Max/MSP. 1,000+ concurrent participants, <2ms P95 latency. Docker/Redis/Nginx. Open-source.

alchemical-synthesizer — Signal-Flow Architecture for Non-Deterministic Logic
Compositional signal-flow system bridging agentic intent and execution reality. Typed signal routing, validation gates, and deterministic state resolution for non-deterministic AI outputs.

EXPERIENCE

Staff Software Engineer & Durable AI Systems — 2020 — Present
ORGANVM

- **Durable State Machines:** 113-repository system with formal promotion pipelines, forward-only progression constraints, and validated transitions preventing silent corruption across 8 GitHub organizations.
- **Multi-Agent Orchestration:** Agentic frameworks with stateful retries, checkpointing, memory management, and self-correction hooks that recover from non-deterministic failures autonomously.
- **SDK & Developer Experience:** 32 agent definitions with typed inputs, validation gates, and formal skill registries. 739K words of documentation across 48 published essays on AI orchestration methodology.
- **CI/CD Infrastructure:** 128 GitHub Actions workflows (18 reusable templates), 104 CI/CD pipelines, automated dependency validation with zero-violation enforcement.

Instructor — Composition & Writing — 2015 — Present
Miami Dade College, Keiser University, Nova Southeastern, FAU, Young Harris, Tarrant County, Broward, Palm Beach State

- 100+ courses across 8 institutions; 2,000+ students. Technical communication, systems-thinking pedagogy, and structured feedback design.

Digital Marketing Manager — 2023 — 2024
Miami Dade College Foundation (nonprofit)

- Rebuilt foundation website; designed data-driven fundraising campaigns increasing donor engagement by 32%. End-to-end ownership from infrastructure through analytics.

Multimedia Specialist — 2011 — 2020
AJP Media Arts — clients incl. Boca Ballet Theatre, Forward Funding, Arts in Boca

- Video, audio, and visual production pipelines for nonprofits and arts organizations; 17.5M+ views, $2M in fundraising.

EDUCATION

M.F.A. Creative Writing, Florida Atlantic University — 2015 — 2018
B.A. English Literature, CUNY College of Staten Island — 2010 — 2014

CERTIFICATIONS

Full-Stack Developer, Meta (2024) · UX Design, Google (2023) · Digital Marketing, Google (2023) · Project Management, Google (2023)